



AJAY PRASATH P 2024-CSE ▾

A2**Started on** Wednesday, 29 October 2025, 10:37 AM**State** Finished**Completed on** Wednesday, 29 October 2025, 11:20 AM**Time taken** 42 mins 48 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Given two strings find the length of the common longest subsequence(need not be contiguous) between the two.

Example:

s1: ggtabe

s2: tgatasb

s1		a	g	g	t	a	b	
s2		g	x	t	x	a	y	b

The length is 4

Solveing it using Dynamic Programming

For example:

Input	Result
aab	2
azb	

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2  #include <string.h>
3
4  #define MAX 100
5
6  int max(int a, int b) {
7      return (a > b) ? a : b;
8  }
9
10 int main() {
11     char s1[MAX], s2[MAX];
12     int dp[MAX][MAX];
13
14     scanf("%s", s1);
15     scanf("%s", s2);
16
17     int m = strlen(s1);
18     int n = strlen(s2);
19
20     for (int i = 0; i <= m; i++) {
21         for (int j = 0; j <= n; j++) {
22             if (i == 0 || j == 0)
23                 dp[i][j] = 0;
24             else if (s1[i - 1] == s2[j - 1])
25                 dp[i][j] = dp[i - 1][j - 1] + 1;
26             else
27                 dp[i][j] = max(dp[i - 1][j], dp[i][j - 1]);
28         }
29     }
30
31     printf("%d\n", dp[m][n]);
32     return 0;
33 }
```

	Input	Expected	Got	
✓	aab azb	2	2	✓
✓	ABCD ABCD	4	4	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)